Neeraj Prasad

FPGA Design Engineer

RESUME_1487

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Total Experience: 6 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 6 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Intel Quartus, Python, Xilinx Vivado, DSP, SPI, AXI4, VHDL, Timing Closure, Verilog, UART, FPGA, ILA

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

Senior FPGA Design Engineer | WillowGrid Labs | Jan 2023 - Present

- Improved maximum operating frequency by 37% after timing-path optimization
- Lowered end-to-end data-path latency by 41% using pipelined logic
- Reduced FPGA resource utilization by 32% through arithmetic sharing
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate FPGA Design Engineer | ZenithTrail Technologies | Jul 2020 - Dec 2022

- Reduced FPGA resource utilization by 24% through arithmetic sharing
- Verified 327 protocol corner cases with repeatable simulation suites

PROJECTS

Camera Data Bridge | DSP, FPGA, VHDL, Timing Closure

- Designed AXI streaming logic, frame buffers and clock-domain boundaries.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

High-Speed Data Logger | Python, AXI4, Xilinx Vivado, UART

- Created DMA and buffering logic with ILA debugging hooks.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Pragati Institute of Applied Sciences, 2020
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

Timing Closure Training (2026)